

PASSIVE 70W TESLA T4: DYNAMIC THERMAL THROTTLING VS OCCUPANCY PACING

Comparison of Unconstrained 100% Occupancy (DVFS Thermal Collapse) vs 25% Paced Boost Lock (Lemma 5)

UNCONSTRAINED OCCUPANCY (100% / 1024 THREADS/SM)

1. Maximum Warps Active 32 Warps / SM
2. Dynamic Power Spikes 78.4W (> 70W TDP)
3. Hardware SW_POWER_CAP (0x0004) Throttling Tripped
4. Frequency Thermal Collapse 1590 → 950 MHz (-40%)

PACED OCCUPANCY (25% / 256 THREADS/SM LAUNCH BOUNDS)

1. Paced Launch Bounds `__launch_bounds__(256, 1)`
2. Thermal Stabilization 50.36W (< 70W Ceiling)
3. Zero Throttling Flag `SW_POWER_CAP = 0`
4. Continuous Boost Lock 1590 MHz Locked (1.47x Gain)